

Module 4

Circuit Analysis in the S Domain

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Transfer function

The transfer function $H(s)$ is the ratio of the output response $Y(s)$ to the input excitation $X(s)$, assuming all initial conditions are zero.

$$H(s) = \frac{Y(s)}{X(s)}$$

$$h(t) = \mathcal{L}^{-1}[H(s)]$$

$$H(s) = \text{Voltage gain} = \frac{V_o(s)}{V_i(s)}$$

$$H(s) = \text{Current gain} = \frac{I_o(s)}{I_i(s)}$$

$$H(s) = \text{Impedance} = \frac{V(s)}{I(s)}$$

$$H(s) = \text{Admittance} = \frac{I(s)}{V(s)}$$

The term $h(t)$ represents the unit impulse response—it is the time-domain response of the network to a unit impulse.

Once we know the impulse response $h(t)$ of a network, we can obtain the response of the network to any input signal in the s-domain

$$Y(s) = H(s)X(s)$$

Transfer function

2. The output of a linear system is $y(t) = 10 e^{-t} \cos 4t u(t)$ when the input is $x(t) = e^{-t} u(t)$.

Find the transfer function of the system and its impulse response.

$$X(s) = \frac{1}{s + 1} \quad \text{and} \quad Y(s) = \frac{10(s + 1)}{(s + 1)^2 + 4^2}$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{10(s + 1)^2}{(s + 1)^2 + 16} = \frac{10(s^2 + 2s + 1)}{s^2 + 2s + 17}$$

$$H(s) = 10 - 40 \frac{4}{(s + 1)^2 + 4^2}$$

$$h(t) = 10\delta(t) - 40e^{-t} \sin 4t u(t)$$

Transfer function

3. The transfer function of a linear system is $H(s) = \frac{2s}{s + 6}$

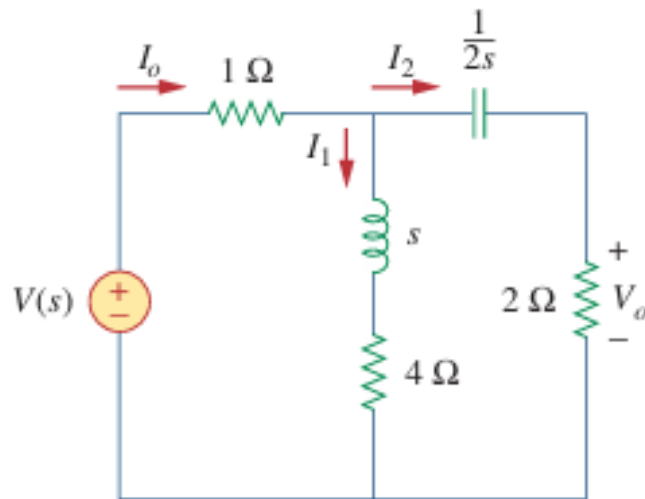
Find the output $u(t)$ due to the input $10e^{-3t} u(t)$ and its impulse response.

Answer: $-20e^{-3t} + 40e^{-6t}, t \geq 0, 2\delta(t) - 12e^{-6t}u(t).$

4. Determine the transfer function $H(s) = V_o(s)/I_o(s)$ of the circuit shown below.

Method:1

By current division,



$$I_2 = \frac{(s + 4)I_o}{s + 4 + 2 + 1/2s}$$

$$V_o = 2I_2 = \frac{2(s + 4)I_o}{s + 6 + 1/2s}$$

$$H(s) = \frac{V_o(s)}{I_o(s)} = \frac{4s(s + 4)}{2s^2 + 12s + 1}$$

Transfer function

Method:2 - ladder method

Let $V_o = 1\text{ V}$

By Ohm's law, $I_2 = V_o/2 = 1/2\text{ A}$

Voltage across $(1/2s + 2)$ is,

$$V_1 = I_2 \left(2 + \frac{1}{2s} \right) = 1 + \frac{1}{4s} = \frac{4s + 1}{4s}$$

This is the same as the voltage across the impedance $(s+4)$,

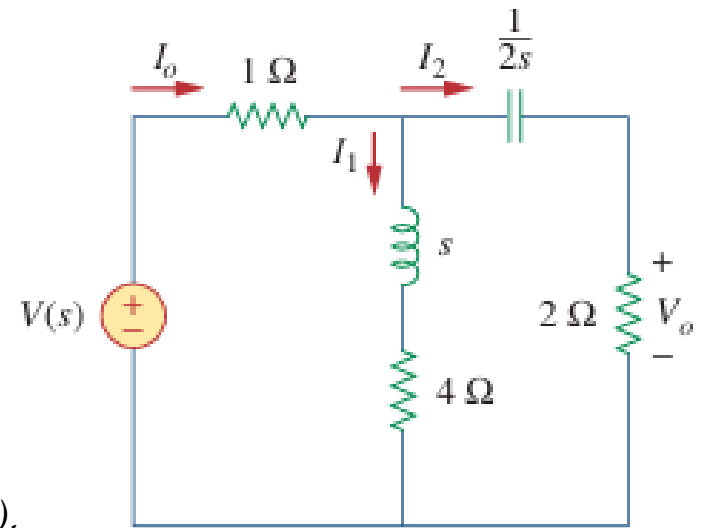
$$H(s) = \frac{V_o}{I_o} = \frac{1}{I_o} = \frac{4s(s + 4)}{2s^2 + 12s + 1}$$

$$I_1 = \frac{V_1}{s + 4} = \frac{4s + 1}{4s(s + 4)}$$

Applying KCL at the top node yields,

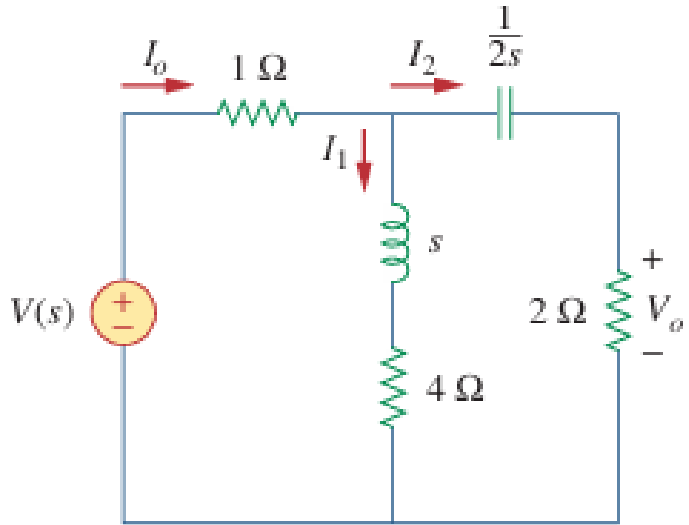
$$I_o = I_1 + I_2 = \frac{4s + 1}{4s(s + 4)} + \frac{1}{2} = \frac{2s^2 + 12s + 1}{4s(s + 4)}$$

$$H(s) = \frac{V_o}{I_o} = \frac{1}{I_o} = \frac{4s(s + 4)}{2s^2 + 12s + 1}$$



Transfer function

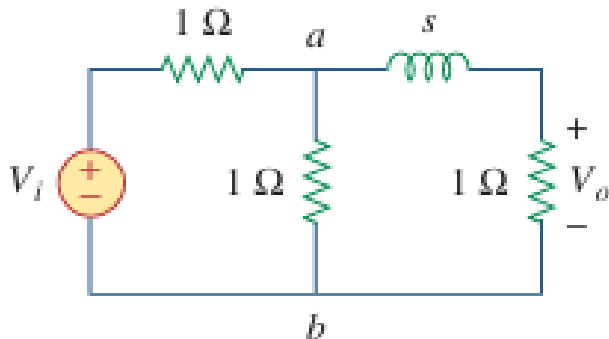
5. Find the transfer function $H(s) = I_1(s)/I_0(s)$ in the circuit.



Answer:

$$H(s) = \frac{I_1}{I_0} = \frac{2 + 1/2s}{s + 4 + 2 + 1/2s} = \frac{4s + 1}{2s^2 + 12s + 1}$$

6. For the s-domain circuit, find: (a) the transfer function $H(s) = V_o(s)/V_i(s)$ (b) the impulse response, (c) the response when $v_i(t) = u(t)$ V (d) the response when $v_i(t) = 8 \cos 2t$ V.



Using voltage division,

$$V_o = \frac{1}{s + 1} V_{ab}$$

However,

$$V_{ab} = \frac{1 \parallel (s + 1)}{1 + 1 \parallel (s + 1)} V_i = \frac{(s + 1)/(s + 2)}{1 + (s + 1)/(s + 2)} V_i$$

Transfer function

$$V_{ab} = \frac{s+1}{2s+3} V_i$$

$$V_o = \frac{V_i}{2s+3}$$

a. $H(s) = \frac{V_o}{V_i} = \frac{1}{2s+3} \qquad H(s) = \frac{1}{2} \frac{1}{s + \frac{3}{2}}$

b. $h(t) = \frac{1}{2} e^{-3t/2} u(t)$

c. when $v_i(t) = u(t)$ $V_i(s) = 1/s$

$$V_o(s) = H(s)V_i(s) = \frac{1}{2s(s + \frac{3}{2})} = \frac{A}{s} + \frac{B}{s + \frac{3}{2}}$$

Transfer function

$$A = sV_o(s) \Big|_{s=0} = \frac{1}{2(s + \frac{3}{2})} \Big|_{s=0} = \frac{1}{3}$$

$$B = \left(s + \frac{3}{2}\right) V_o(s) \Big|_{s=-3/2} = \frac{1}{2s} \Big|_{s=-3/2} = -\frac{1}{3}$$

$$V_o(s) = \frac{1}{3} \left(\frac{1}{s} - \frac{1}{s + \frac{3}{2}} \right)$$

$$v_o(t) = \frac{1}{3}(1 - e^{-3t/2})u(t) \text{ V}$$

d. when $v_i(t) = 8 \cos 2t \text{ V}$ $V_i(s) = \frac{8s}{s^2 + 4}$

$$V_o(s) = H(s)V_i(s) = \frac{4s}{(s + \frac{3}{2})(s^2 + 4)}$$

$$= \frac{A}{s + \frac{3}{2}} + \frac{Bs + C}{s^2 + 4}$$

$$A = \left(s + \frac{3}{2}\right) V_o(s) \Big|_{s=-3/2} = \frac{4s}{s^2 + 4} \Big|_{s=-3/2} = -\frac{24}{25}$$

Transfer function

To get B and C,

$$4s = A(s^2 + 4) + B\left(s^2 + \frac{3}{2}s\right) + C\left(s + \frac{3}{2}\right)$$

$$0 = 4A + \frac{3}{2}C \quad \Rightarrow \quad C = -\frac{8}{3}A$$

$$4 = \frac{3}{2}B + C$$

$$0 = A + B \quad \Rightarrow \quad B = -A$$

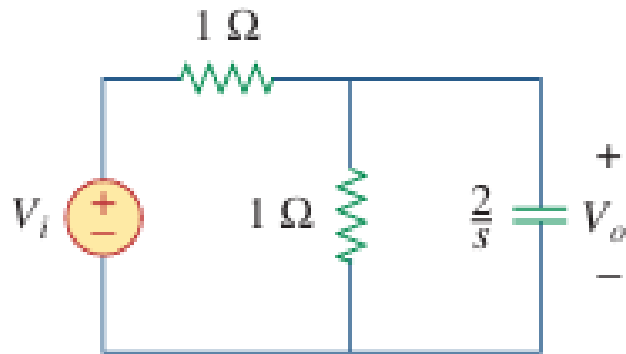
B = 24/25, and C = 64/25,

$$V_o(s) = \frac{-\frac{24}{25}}{s + \frac{3}{2}} + \frac{24}{25} \frac{s}{s^2 + 4} + \frac{32}{25} \frac{2}{s^2 + 4}$$

$$v_o(t) = \frac{24}{25} \left(-e^{-3t/2} + \cos 2t + \frac{4}{3} \sin 2t \right) u(t) \text{ V}$$

Transfer function

7. for the circuit shown below, find (a) the transfer function $H(s) = V_o(s)/V_i(s)$ (b) the impulse response, (c) the response when $v_i(t) = u(t)$ V (d) the response when $v_i(t) = 8 \cos 2t$ V.



Answer: (a) $2/(s + 4)$, (b) $2e^{-4t}u(t)$, (c) $\frac{1}{2}(1 - e^{-4t})u(t)$ V,
(d) $3.2(-e^{-4t} + \cos 2t + \frac{1}{2} \sin 2t)u(t)$ V.

Next class: Passive Filters